

PHYSICS
(SCIENCE PAPER – 1)

Maximum Marks: 80

Time allowed: Two hours

1. *Answers to this Paper must be written on the paper provided separately.*
 2. *You will **not** be allowed to write during first 15 minutes.*
 3. *This time is to be spent in reading the question paper.*
 4. ***The time given at the head of this Paper is the time allowed for writing the answers.***
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5. ***Section A is compulsory. Attempt any four questions from Section B.***
 6. *The intended marks for questions or parts of questions are given in brackets [].*

Instruction for the Supervising Examiner

*Kindly read aloud the Instructions given above to all the candidates present in the
Examination Hall.*

This Paper consists of 16 printed pages.

SECTION A (40 Marks)

(Attempt *all* questions from this *Section*.)

Question 1

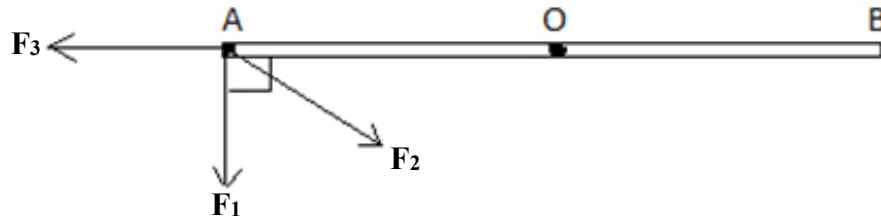
Choose the correct answers to the questions from the given options.

[15]

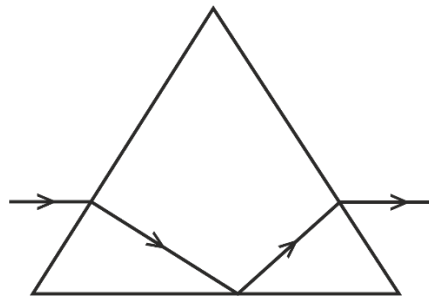
(Do not copy the questions, write the correct answers only.)

- (i) The energy transformation taking place in a **Thermocouple** is:
- (a) Heat energy to electrical energy
 - (b) Heat energy to mechanical energy
 - (c) Electrical energy to heat energy
 - (d) Electrical energy to mechanical energy
- (ii) The unit of the physical quantity expressed by equation $\frac{p^2}{2m}$ is: (*p is momentum and m is mass*)
- (a) newton
 - (b) joule
 - (c) watt
 - (d) joule per kilogram
- (iii) A block and tackle system has **two** pulleys in the **movable** block. In an ideal condition if $L = 5T$, then the number of pulleys in the fixed block is equal to:
- (*L is the load and T is the tension*)
- (a) 2
 - (b) 3
 - (c) 4
 - (d) 5

- (iv) If three equal forces F_1 , F_2 and F_3 are acting at the end A of the rod AB, pivoted at the center O, as shown in the diagram given below, then the **moment of force** is:



- (a) least for F_2 and maximum for F_1
 (b) least for F_1 and maximum for F_3
 (c) least for F_3 and maximum for F_1
 (d) least for F_2 and maximum for F_3
- (v) The **net** angle of deviation produced by the prism in the diagram shown below is _____.



- (a) 0°
 (b) 60°
 (c) 90°
 (d) 180°

- (vi) A crack in a glass vessel often shines, like a mirror, in sunlight. Which of the following **phenomenon** of light is responsible for this?
- (a) reflection
 - (b) refraction
 - (c) dispersion
 - (d) total internal reflection
- (vii) The **magnification** produced by a lens is +2. Then the image is:
- (a) real and magnified
 - (b) virtual and diminished
 - (c) real and diminished
 - (d) virtual and magnified
- (viii) Identify the radiation that scatters the **least**.
- (a) Radio waves
 - (b) Microwaves
 - (c) X-rays
 - (d) Gamma rays
- (ix) **Assertion (A):** The buzzing of a mosquito is shriller than the roar of a lion.
Reason (R): The frequency of sound produced by a mosquito is greater than the frequency of sound produced by the roar of a lion.
- (a) (A) is true but (R) is false.
 - (b) (A) is false but (R) is true.
 - (c) Both (A) and (R) are true and (R) is the correct explanation of (A).
 - (d) Both (A) and (R) are true but (R) is not the correct explanation of (A).

- (x) A straight wire of resistance **R** is bent in a U shape. Its resistance is:
- (a) $\frac{3}{4}R$
 - (b) $\frac{R}{2}$
 - (c) R
 - (d) 2 R
- (xi) Which of the following is **NOT** a characteristic of a fuse wire?
- (a) It has high resistance.
 - (b) It has high conductivity.
 - (c) It is an alloy of lead and tin.
 - (d) It has a low melting point.
- (xii) Identify the **correct** statement with respect to the change of phase of a substance.
- (a) Kinetic energy of the molecules remains the same.
 - (b) Potential energy of the molecules remains the same.
 - (c) Both potential and kinetic energies of the molecules increase.
 - (d) Both potential and kinetic energies of the molecules remain the same.
- (xiii) **1 calorie** is the amount of heat required to raise the temperature of:
- (a) 1 kg of water through 1 K
 - (b) 1 g of water through 1 K
 - (c) 1 kg of water through 1°C
 - (d) 1 g of water through 10°C

- (xiv) A current carrying conductor experiences a force when placed **normal** to a magnetic field. The force experienced **does not depend** on the:
- (a) mass of the conductor
 - (b) current through the conductor
 - (c) strength of the magnetic field
 - (d) length of the conductor
- (xv) When the nuclear radiation is subjected to a magnetic field, the **direction of deflection** is governed by:
- (a) Flemming's Right Hand Rule
 - (b) Flemming's Left Hand Rule
 - (c) Clock Rule
 - (d) Right Hand Thumb Rule

Question 2

- (i) Complete the following by choosing the correct answers from the bracket: [6]
- (a) If the displacement of a load is **greater** than the displacement of the effort, then the velocity ratio of the machine is _____. [*less than one / more than one*].
 - (b) Out of multiple images of an object, formed by a thick plane glass mirror, the _____ [*first / second*] image is the **brightest**.
 - (c) No **negative** temperature is possible on the _____. [*Kelvin scale / Fahrenheit scale*]
 - (d) 250 dB is _____ [*shriller / louder*] than 10 dB.

- (e) In a series combination, across each resistor, the _____ [*current / potential difference*] remains the **same**.
- (f) ${}_{18}\text{Ar}^{40}$ and ${}_{20}\text{Ca}^{40}$ are the examples of _____ [*isobars / isotopes*].
- (ii) Rekha has a metre ruler weighing 200 gf, half (50 cm) made of wood and half made of steel. [2]
- (a) Rekha wants to balance the ruler. To which part (*wooden or steel*) should she keep the knife edge?
- (b) The above metre-rule is to be balanced by placing the knife edge at 50 cm mark using a known weight. **Name the principle** which can be used to calculate the position of the weight.
- (iii) (a) In a **second-class lever**, the ratio of the effort arm to the load arm is _____ (*greater than 1 / less than 1*). [2]
- (b) State *one* advantage of the combination of **one fixed** and **one movable** pulley over a **single movable pulley**.

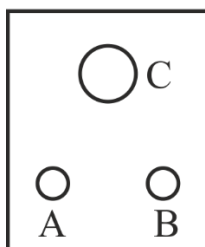
Question 3

- (i) An upright lens is divided horizontally into a number of parts and the diagram given below shows the magnified view of its **lowest part**. [2]



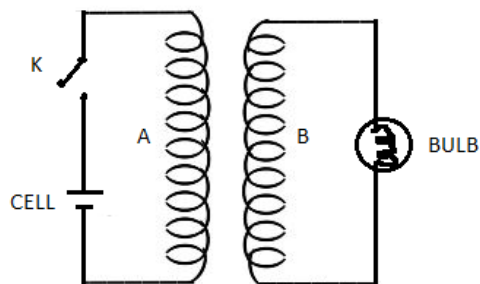
- (a) Identify the lens.
- (b) Justify your answer in (a) above.

- (ii) (a) Why are **thick wires** used to carry current in high power appliances such as electric heaters and radiators? [2]
- (b) Name a device used to protect the electric circuits from overloading and short circuits.
- (iii) The diagram below shows a socket in which the **switch is ON**. [2]



Will the bulb glow if:

- (a) it is connected across B and C?
- (b) it is connected across A and C?
- (iv) The temperatures of two liquids are 80°C and 20°C . When they are mixed in the mass ratio of 8:5, the temperature of the mixture becomes 60°C . Calculate the ratio of their specific heat capacities. (*Neglecting heat lost to the surroundings*) [2]
- (v) In the given figure, a cell is connected to a coil A and a bulb is connected to coil B. [2]



- (a) **Fill in the blank.**
When the key 'K' is closed, then the _____ [*bulb glows continuously / bulb glows only for a moment*].
- (b) Support your answer to (a) above with a suitable reason.

- (vi) A uranium nucleus (${}^{235}_{92}\text{U}$) undergoes **an alpha** emission followed by **2 beta** emissions. [2]

Calculate the atomic number and mass number of the daughter nucleus formed.

- (vii) Two beakers A and B contain kerosene oil and water respectively. The speed of light in water is $2.25 \times 10^8 \text{ms}^{-1}$ and in kerosene oil $2.15 \times 10^8 \text{ms}^{-1}$. [3]

- (a) In which liquid will the angle of refraction be **more**, when light ray strikes the surface of both liquids at the same angle of incidence?
- (b) Calculate the ratio of refractive indices of water to kerosene oil.
- (c) If liquid A is heated, what change will you expect in the path of the refracted ray?

SECTION B (40 Marks)

*(Attempt **any four** questions from this **Section**.)*

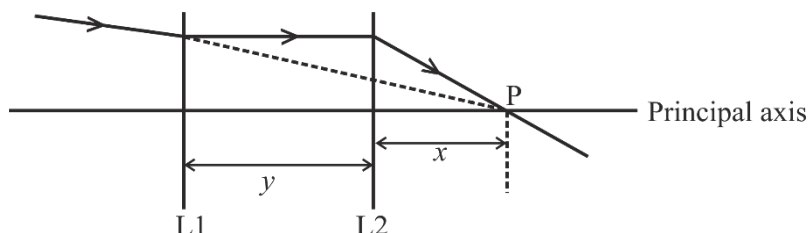
Question 4

- (i) Rohit is travelling by a car through a highway on a hot summer afternoon. The road in front of him **appears momentarily wet** as shown in the picture below. [3]



- (a) Name and define the phenomenon responsible for this.
- (b) Provide another example that demonstrates the same phenomenon.

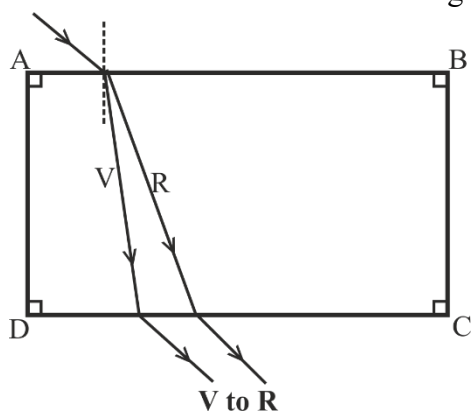
- (ii) In the diagram given below, a concave (L1) and a convex (L2) lens of focal lengths 20 cm and 12 cm respectively are placed parallel to each other such that they have a common principal axis. Study the path of the ray through the lenses. [3]



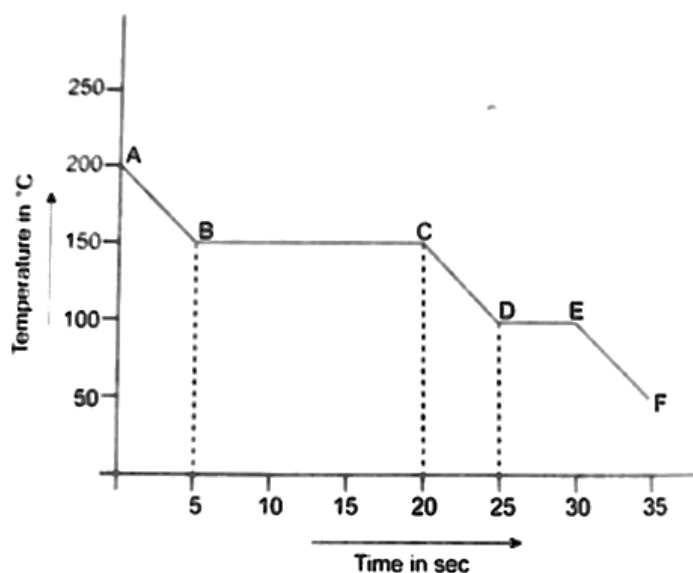
- (a) Calculate the lengths of 'x' and 'y'.
- (b) Define **point P** with respect to the **concave** lens (L1).
- (iii) (a) In an experiment to find the refractive index of glass, if yellow light is replaced by green light, then how will the **refractive index** of the glass change? Give a reason in support of your answer. [4]
- (b) What is the value of angle of **deviation**, when light ray falls **normally** on the surface during refraction?
- (c) Is it possible to have a **critical angle** of water with respect to glass? (*Write YES or NO*)

Question 5

- (i) The diagram given below shows refraction of white light through a rectangular glass block. [3]



- (a) Name the *two phenomena* taking place at the surface AB.
- (b) With a **reason** state whether one can see these colours when seen from the side DC.
- (ii) The graph given below shows the change in temperature with time, when a substance is cooled. [3]



- (a) State the boiling point of the substance.
- (b) What is the phase change occurring in the region DE?
- (c) State the range of temperature during which the substance is in liquid state.
- (iii) 280 g of ice at 0°C is mixed with 600 g of water at 50°C present in a metal container of mass 300 g. [4]

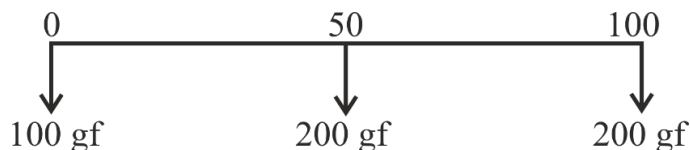
If the final temperature of the mixture is 10°C. Assuming no loss of heat to the atmosphere, calculate the specific heat capacity of the metal container.

(Specific latent heat of ice is 336 Jg^{-1} ,

Specific heat capacity of water is $4.2 \text{ Jg}^{-1}\text{K}^{-1}$)

Question 6

- (i) Observe the diagram of a **uniform** metre scale of mass 200 g given below. [3]
Weights of 200 gf and 100 gf are suspended from its ends. At which marking will the ruler be **balanced**?



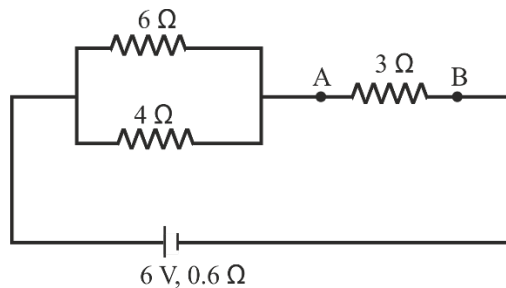
- (ii) A constant force **F** acts on a horizontal uniform ruler AB at point A. [3]
- Case 1. The ruler is pivoted at the end B.
- Case 2. Ruler is not pivoted.
- (a) In which case is the moment of force greater? (*Case 1 or Case 2*)
- (b) Give a reason for your answer in (a) above.
- (c) In which case does the ruler exhibit both translational and rotational motion simultaneously? (*Case 1 or Case 2*)
- (iii) (a) **Derive** a relation between mechanical advantage (MA), velocity ratio (VR) and efficiency (η). [4]
- (b) A stapler of Class III lever is used to staple papers.
1. Is this lever a speed multiplier or a force multiplier?
 2. Give a reason for your answer to (1).

Question 7

- (i) A block and tackle system of pulleys has a Velocity Ratio (VR) = 5. Draw a labelled diagram of the system indicating clearly the direction of tension, load and effort. [3]
- (ii) It is observed during an earthquake, that the short period seismic waves damage low rise buildings while long period waves damage high rise buildings. [3]
- (a) Name the phenomenon responsible for the damage of the buildings.
- (b) Why do long period seismic waves damage high rise buildings?
- (iii) A person standing between two vertical cliffs produces a sound and hears the first echo after 3 s. If the distance between the cliffs is 1760 m, then after how much time will he hear the second echo? (*The speed of sound = 320 ms^{-1}*) [4]

Question 8

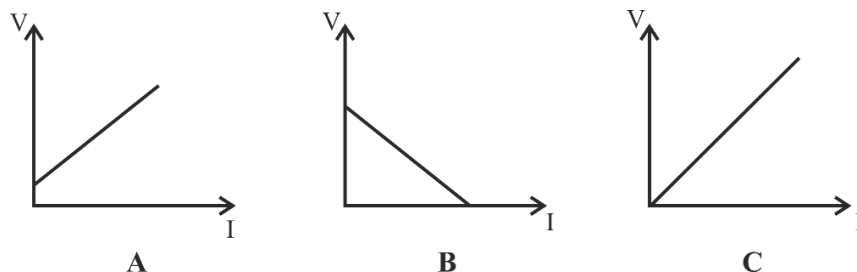
- (i) Observe the circuit diagram given below. [3]



Calculate the current:

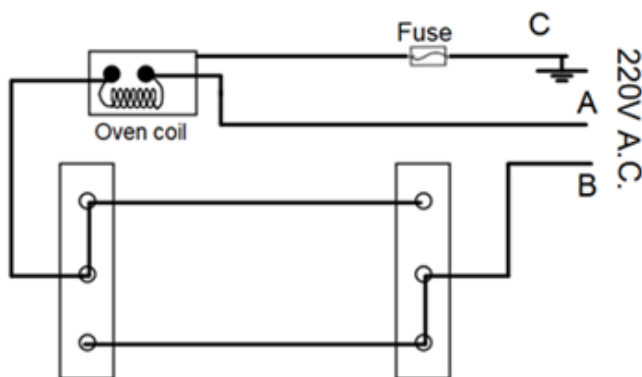
- (a) through 3Ω resistor.
- (b) drawn from the cell when the 3Ω resistance is replaced by a connecting wire of negligible resistance.

- (ii) (a) Which of the following graphs (A, B or C) shows the correct relation between Potential difference (V) and Current (I) for an **Ohmic resistor**? [3]
Give a reason for your answer.



- (b) Give an example of an Ohmic resistor.

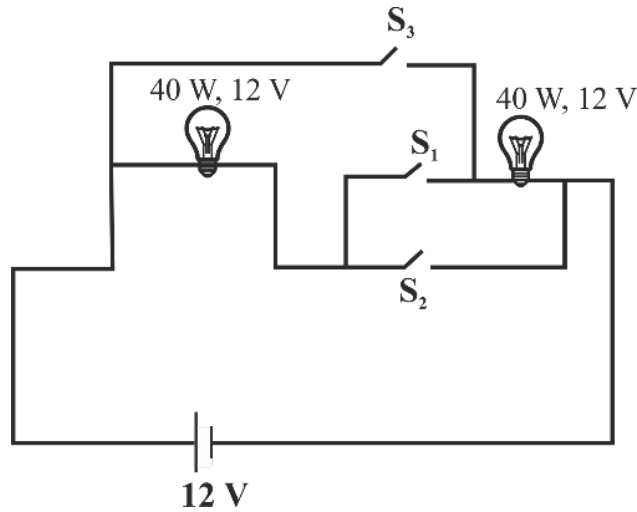
- (iii) (a) Ravi wants to install a fuse to the circuit of an oven and wants to control the oven from two different locations. Shown below is the circuit diagram which he used. [4]



1. Which one of the two, A or B, should be a live wire? Why?
 2. In the event of an overload, will the fuse serve its purpose?
- (b) What does the statement '*A bulb is rated 600W, 220 V*' mean?

Question 9

- (i) The diagram given below shows two bulbs rated 40 W, 12 V each. [3]



- (a) Calculate the resistance of a bulb.
- (b) What is the total power consumed by the circuit:
1. when S_1 is **closed** and S_2 and S_3 are **open**?
 2. when S_1 is **open** and S_2 and S_3 are **closed**?
- (ii) (a) Draw a diagram showing the **magnetic lines of force** around the straight wire [3]
wire carrying current.
- (b) What will be the effect on the magnetic field if the current through the wire **increases**?

(iii) Scientists unearthed an ancient scroll for determining the **age** of a fossil. The **process** involved collecting $^{14}_6\text{C}$ from the fossil and comparing it with a fresh sample. Scientist found that the ratio of $^{14}_6\text{C} : ^{12}_6\text{C}$ **decreased** as compared to the fresh sample. [4]

(a) Name the process.

(b) $^{14}_6\text{C}$ and $^{12}_6\text{C}$ are collectively known as _____.

(c) Why did the ratio decrease in the fossil?

(d) Complete the equation by filling the blanks:

